

Academic and Research Experience

Tongji University

- Research Assistant Mar 2026 - Present
- Ph.D. in Vehicle Engineering (Advisor: Prof. Lu Xiong, Co-advisor: Prof. Jia Hu and Prof. Nan Li) Sep 2019 - Dec 2025
- B.E. in Engineering Mechanics Sep 2014 - Jul 2019

Research Directions

LLM-Enhanced End-to-End Autonomous Driving

Dec 2024 - Present

- Scenario understanding for adaptive reward adjustment during E2E RL training
- Physical feasibility enhanced diffusion planning

Control Granularity Improved RL-based Motion Planning

Dec 2023 - Jun 2025

- Hierarchical RL for integrated decision-making and motion control
- Hybrid action-based RL using parameterized action space

Safety-Enhanced Reinforcement Learning for Autonomous Driving

Sept 2022 - Present

- Prior-knowledge based predictive risk constraint and unsafe behavior correction.
- Multi-Critic mechanism for multi-objective accommodation
- Epistemic uncertainty-based action governor with dynamic control barrier function

Optimization-based Motion Planning

Sept 2021 - Jun 2023

- POMDP-based integrated decision-making and motion planning considering prediction uncertainty
- MPC-based Trajectory planning and tracking control

Selected Publications

Journal & Conference Publications:

- Dequan Zeng, Zhizhao Ni, **Zhuoren Li**^{*}, et al., "Physics-Informed Residual Reinforcement Learning via Expert Prior Knowledge for Safe and Efficient Autonomous Merging," *Neurocomputing*. (accept)
- Bo Leng, Weiqi Zhang, **Zhuoren Li**^{*}, et al., "Expert Knowledge-driven Reinforcement Learning for Autonomous Racing via Trajectory Guidance and Dynamics Constraints," *Green Energy Intell. Transp.*, 100452, 2026. [PDF]
- Zhuoren Li**, Bo Leng, Lu Xiong, Arno Eichberger, Chao Huang and Jia Hu, "Safety Enhanced Reinforcement Learning for Autonomous Driving: Dare to Make Mistakes to Learn Faster and Better," *IEEE Trans. Intell. Transp. Syst.*, 2026. [PDF]
- Guizhe Jin, **Zhuoren Li**, Bo Leng, et al. "Hybrid Action Based Reinforcement Learning for Multi-Objective Compatible Autonomous Driving," *IEEE Trans. Neural Netw. Learn. Sys.*, 2026. [PDF]
- Bo Leng, Ran Yu, Chengen Tu, Lu Xiong, Arno Eichberger and **Zhuoren Li**^{*}, "Seamless Overtaking Maneuvers for Automated Driving: Integrated Motion Planning Based on Hybrid Model Predictive Control," *IEEE Trans. Ind. Electron.*, 2026. [PDF]
- Zhiwen Chen, Hanming Deng, **Zhuoren Li**^{*}, Huanxi Wen, Guizhe Jin, Ran Yu and Bo Leng^{*}, "HCRMP: A LLM-Hinted Contextual Reinforcement Learning Framework for Autonomous Driving," *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2025. [PDF]
- Ran Yu, **Zhuoren Li**^{*}, Lu Xiong, et al., "Uncertainty-Aware Safety-Critical Decision and Control for Autonomous Vehicles at Unsignalized Intersections," *IEEE Intell. Transp. Syst. Conf. (ITSC)*, 2025. [PDF]
- Guizhe Jin, **Zhuoren Li**, Bo Leng, Ran Yu, Lu Xiong and Chen Sun, "Multi-Timescale Hierarchical Reinforcement Learning for Unified Behavior and Control of Autonomous Driving," *IEEE Robot. Autom. Lett.*, 10(12): 12772-12779, 2025. [PDF]
- Bo Leng, **Zhuoren Li**^{*}, Ming Liu, Ce Yang, Yi Luo, Amir Khajepour and Lu Xiong, "Multi-Mode Evasion Assistance Control Method considering Human Driver Operation," *Chin. J. Mech. Eng.*, 2025, 38: 102. [PDF]
- Lu Xiong⁺, **Zhuoren Li**, Danyang Zhong, Puhang Xu and Chen Tang, "Rule-Guidance Reinforcement Learning for Lane Change Decision-making: A Risk Assessment Approach," *Chin. J. Mech. Eng.*, 2025, 38:30. [PDF]
- Ruolin Yang, **Zhuoren Li**, Bo Leng, Lu Xiong and Xin Xia, "Convergent Harmonious Decision: Lane Changing in a more Traffic Friendly Way," *IEEE Trans. Intell. Transp. Syst.*, 26(11): 20334-20347, 2025. [PDF]
- Zhuoren Li**, Jia Hu, Bo Leng, Lu Xiong and Zhiqiang Fu, "An Integrated of Decision Making and Motion Planning Framework for Enhanced Oscillation-Free Capability," *IEEE Trans. Intell. Transp. Syst.*, 25(6): 5718-5732, 2024. [PDF]
- Zhuoren Li**, Guizhe Jin, Ran Yu, Bo Leng and Lu Xiong, "Interaction-Aware Deep Reinforcement Learning Approach Based on Hybrid Parameterized Action Space for Autonomous Driving," *SAE Intell. Connected Veh. Symposium*, 2024. [PDF]
- Zhuoren Li**, Lu Xiong, Bo Leng, Puhang Xu and Zhiqiang Fu, "Safe Reinforcement Learning of Lane Change Decision Making with Risk-Fused Constraint," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, 2023, pp. 1313-1319. [PDF]
- Zhuoren Li**, Lu Xiong Bo Leng. "A Unified Trajectory Planning and Tracking Control Framework for Autonomous Overtaking Based on Hierarchical MPC," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, 2022, pp. 937-944. [PDF]

Submitted & In Progress:

- Zhuoren Li**, Yi Zhong, Weiqi Zhang, et al. "Comparison-Based Ordinal Learning for Proactive Driving Risk Assessment." *Accident Analysis & Prevention* (under review) [arXiv]
- Zhuoren Li**, Guizhe Jin, Ran Yu, et al. "A Survey of Reinforcement Learning-Based Motion Planning for Autonomous Driving: Lessons Learned from a Driving Task Perspective." *IEEE Trans. Intell. Transp. Syst.* (under review) [arXiv]
- Bo Leng, Ran Yu, **Zhuoren Li**^{*} et al., "Risk-Aware Reinforcement Learning for Autonomous Driving: Improving Safety When Driving through Intersection," *Eng. Appl. Artif. Intel.* (under review) [arXiv]

^{*} corresponding author, ⁺ First author of doctoral supervisor

Selected Projects

- Interactive Learning-based Decision Control for Intelligent Vehicles with Roadside Information Enhancement** Apr 2025 - Present
Chongqing Changan Automobile Co., Ltd., Role: Student Technical Director
- Comprehensive task management.
 - Designed an Interaction-aware RL motion planning algorithm combining with game theory guidance.
- High-Mobility Motion Planning and Control Research for Chassis-by-wire All-terrain Unmanned Vehicle with Hybrid-steering** Jan 2024 - Present
National Natural Science Foundation of China, Role: Student Technical Director
- Comprehensive task management.
 - Designed an environmental risk characterization algorithm considering negative obstacles.
 - Led the Development of a motion planning approach combining hybrid A* and optimization-based methods.
- Key Technology of Perception and Control in Cooperative Vehicle-Infrastructure System for Urban Public Transportation** Jan 2023 – Dec 2025
National Key Research and Development Program of China, Role: Core Participant
- Developed the decision-making and motion planning algorithms for cruise driving and lane change functions in vehicle side, based on the recommended speed from the road side.
- Development of Evasion Assistance Algorithm for Emergency Collision Avoidance based on Steer-by-Wire System** Jul 2022 – Jul 2024
Shanghai Automotive Industry Science and Technology Development Foundation, Role: Student Technical Director
- Comprehensive task management.
 - Led the development of a motion control algorithm for emergency collision avoidance.
 - Led the simulation validation in Carmaker/Simulink, steer-by-wire system modification on the real vehicle and ground test based on Audesse ECU.
- Binary Mixed Traffic Behavior Characteristics and Collaborative Paradigm** Aug 2021 – Jul 2024
Science and Technology Commission of Shanghai, Role: Core Participant, Role: Work Package Leader
- Motion planning and control of connected automated vehicle according to the road-side guidance.
- Research on Vehicle-Road Cooperative Control for Intelligent Public Transportation System** Sep 2021 – Aug 2022
Shanghai Research Institute for Intelligent Autonomous Systems, - Role: Student Technical Director
- Comprehensive task management.
 - Led the development of a signal control algorithm and a vehicle speed optimization algorithm.
- Development and Application of Automatic Valet Parking System** Feb 2020 – Mar 2021
Nanchang Automotive Institute of Intelligence & New Energy, Role: Student Technical Director
- Comprehensive task management.
 - Developed an integrated module for hardware and software communication.
 - Developed a parking path planning algorithm based on geometric configuration.
 - Developed a Stanley-based path tracking algorithm and a feedforward PID-based speed tracking algorithm.
- China Future Challenge of Intelligent Vehicles** Mar 2020 – Nov 2020
Institute of Intelligent Vehicles, Tongji University, Role: Major developer
- Developed a system integration framework inspired by the Baidu Apollo.
 - Developed a search-based path planning algorithm for static obstacle avoidance.
 - Developed a parking path planning algorithm based on hybrid A*.
- Funding/Grant Proposal Writing Experience** Jan 2020 – Mar 2025
- National Natural Science Foundation of China (NSFC): Excellent Young Scholars Fund, 2025; Key Program, 2024; General Program, 2023; Distinguished Young Scholars Fund, 2022; Young Scholars Grant, 2020.
 - Shanghai Municipal People's Government (SMPG), Oriental Excellence Program Youth Project, 2024;
 - Shanghai Science and Technology Progress First Award, 2022;
 - Ministry of Science and Technology, PRC, National Key Research and Development Program of China, 2021;

Honors and Awards

- SAE International Outstanding Technical Paper Award, SAE ICVS 2024.
- High-Level Academic Poster Award, China SAE Doctoral Student Academic Forum, 2024.
- Outstanding Doctoral Student Scholarship, Tongji University, 2024.
- Vehicle-road-cloud Integrated Autonomous Driving Challenge, Third Prize, 2024.
- Outstanding Individual Award, Institute of Intelligent Vehicles, Tongji University, 2022, 2023, 2024.
- Outstanding Projects Award, Institute of Intelligent Vehicles, Tongji University, 2024.
- World Artificial Intelligence Conference AI Driving Simulation Competition, Third Prize in the University Challenge Competition, 2022

Academic Services

Reviewer

- **Journal Reviewer:** *IEEE Transactions on Intelligent Transportation Systems (TITS)*, *IEEE Transactions on Vehicular Technology (TVT)*, *IEEE Transactions on Intelligent Vehicles (TIV)*, *IEEE Transactions on Transportation Electrification (TTE)*, *IEEE Robotics and Automation Letters (RAL)*, *Engineering Applications of Artificial Intelligence (EAAI)*, *Journal of Intelligent Transportation Systems (JITS)*, *IET Intelligent Transport Systems*, *Journal of Field Robotics*.
 - **Conference Reviewer:** *IV*, *ITSC*, *CVCI*, *SAE WCX*, *CCTA*
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Mentoring (serving as the Student Director of the Intelligent Decision Research Group at TJU-IIV since 2021, mentored 5 Ph.D.students, 13 master's students, and several undergraduate students.)

Ph.D. Students:

- 2025-present: **WeiQi Zhang**, E2E RL racing; Baoyun Wang, VLA-based E2E control
- 2023-present: **Zhiwen Chen**, LLM-enhanced RL for AD.
- 2022-present: **Peiyuan Fang**, Motion Planning under Off-road Environment; **Xinrui Zhang**, Cloud-Vehicle Cooperative Planning.

Master Students:

- 2024-present: **Ran Yu**, Trajectory Prediction and Safe-RL; **Zhizhao Ni**, LfD-based RL for Merging in Congested Traffic.
- 2023-present: **Guizhe Jin**, Multi-objective Compatible RL; **Zhou Sun**, Motion Planning under Off-road Environment.
- 2022-present: **Yuqin Qi**, Learning-based MPC for Motion Control; **Yu Che**, Cloud-Vehicle Cooperative Tracking Control.
- 2021-2024: **Ruolin Yang**, Harmony-enchanced RL; **Encheng Tu**, Hybrid MPC Motion Planning for Autonomous Overtaking; **Yizhuo Guan**, Prediction and Control for Emergency Evasion.
- 2020-2023: **Gesong Shi**, Cooperative Control for Transit Priority.
- 2019-2022: **Puhang Xu**, Safe-RL Decision-Making; **Hongyu Xiao**, POMDP Motion Planning; **Zixuan Qian**, SMPC Motion Planning; **Jie Gao**: Global Planning.